STATISTICS MASTER'S COMPREHENSIVE EXAMINATION

3 December 2022

Instructions: Attempt any five out of the following six questions. Specify which five you would like to have graded. Students are allowed calculators, laptops, books and notes – no internet use and no cell phones out.

Problem 1. Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be i.i.d. samples with $X_i \sim N(0,1)$ and $Y_i|X_i = x \sim N(x^2\theta, 1)$.

(1) Find the MLE $\hat{\theta}_n$ of θ.
Answer: The joint density of (X, Y) is $f_\theta(x, y) = \exp(-x^2/2 - (y - x^2\theta)^2/2)/2\pi$, so the log likelihood function is

$$\ell_n(\theta) = -\frac{1}{2}\sum_{i=1}^{n} X_i^2 - \frac{1}{2}\sum_{i=1}^{n}(Y_i - X_i^2\theta)^2 - n\log(2\pi).$$

Hence the MLE is solved by the zero of the derivative the likelihood function

$$\ell_n'(\theta) = -2\sum_{i=1}^{n} X_i^2(Y_i - X_i^2\theta) = 0,$$

i.e.

$$\hat{\theta}_n = \frac{\sum_{i=1}^{n} X_i Y_i}{\sum_{i=1}^{n} X_i^4}.$$

(2) Find the limiting distribution of $\sqrt{n}(\hat{\theta}_n - \theta)$.
Answer: The Fisher information is

$$I(\theta) = -E_\theta \frac{\mathrm{d}^2}{\mathrm{d}\theta^2}\log f_\theta(X, Y) = E_\theta X^4 = 3.$$

Hence

$$\sqrt{n}(\hat{\theta}_n - \theta) \to_d N(0, 1/3).$$

Problem 2. Lord Rayleigh was one of the earliest scientists to study the density of nitrogen. In his studies, he noticed something peculiar. The nitrogen densities produced from chemical compounds tended to be smaller than the densities of nitrogen produced from the atmosphere. Lord Rayleigh's measurements 18 are given in the following table. These measurements correspond to the mass of nitrogen filling a flask of specified volume under specified temperature and pressure.

Compound Chemical	Atmosphere
2.30143	2.31017
2.29890	2.30986
2.29816	2.31010
2.30182	2.31001
2.29869	2.31024
2.29940	2.31010
2.29849	2.31028
2.29889	2.31163
2.30074	2.30956
2.30054	

(1) Is there sufficient evidence to indicate a difference in the mean mass of nitrogen per flask for chemical compounds and the atmosphere? What can be said about the p-value associated with your test?
Answer: The hypotheses are $H_0 : \mu_1 - \mu_2 = 0$ *vs.* $H_1 : \mu_1 - \mu_2 \neq 0$*, where* $\mu_1 =$ *mean nitrogen density for chemical compounds and* $\mu_2 =$ *mean nitrogen density for air. Then the value of t-test can be computed to be* $|t| = 22.17$ *with 17-degrees of freedom. The p-value is extremely small (and uninteresting to explicitly write down).*

(2) Find a 95% confidence interval for the difference in mean mass of nitrogen per flask for chemical compounds and air.
Answer: The 95% CI for $\mu_1 - \mu_2$ *is (–.001151, –0.000951) that does not contain 0, so there is evidence that the mean densities are different.*

Problem 3. Suppose we plan to study the effects of two vaccines for COVID-19, vaccine A and vaccine B. We collect n observations, denoted by $(y_1, \cdots, y_n)$, from the group taking vaccine A and m observations, denoted by $(y_{n+1}, \cdots, y_{n+m})$, from the group taking vaccine B, and k observations from the control group (neither A nor B), denoted by $(y_{n+m+1}, \cdots, y_{n+m+k})$.

(1) Denote the overall mean effect by μ, vaccine A group effect by α_1, vaccine B group effect by α_2, and the control group effect by α_3. Write down a regression model to study the effects of vaccinations.
Answer: $y_i = \mu + \alpha_1 + \varepsilon_i$ *for* $i = 1, 2, \ldots, n$, $y_i = \mu + \alpha_2 + \varepsilon_i$ *for* $i = n+1, \ldots, n+m$, $y_i = \mu + \alpha_3 + \varepsilon_i$ *for* $i = n+m+1, \ldots, n+m+k$, *where* ε_i *are iid* $N(0, \sigma^2)$ *for* $i = 1, \ldots, n+m+k$.

(2) What is the corresponding model matrix for (1)?

$$\textit{Answer: } y = X\beta+\varepsilon \textit{ where } y = \begin{pmatrix} y_1 \\ \dots \\ y_{n+m+k} \end{pmatrix}, X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & 0 & 0 & 1 \end{pmatrix},$$

$$\beta = \begin{pmatrix} \mu \\ \alpha_1 \\ \alpha_2 \\ \alpha_2 \end{pmatrix}, \textit{ and } \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \dots \\ \varepsilon_{n+m+k} \end{pmatrix},$$

(3) Is it possible to estimate all the parameters, including μ, α_1, α_2, and α_3? If yes, please explain the estimation procedure. If not, please write down the reasons and provide a solution that makes the estimation possible.

$$\textit{Answer: No. } \beta \textit{ is not identifiable because } X^\top X = \begin{pmatrix} n+m+k & n & m & k \\ n & n & 0 & 0 \\ m & 0 & m & 0 \\ k & 0 & 0 & k \end{pmatrix}$$

is not full rank. The first row equals the sum of the next three rows. Make β identifiable by putting adding a constraint line $\mu = 0$ or $\alpha_1 + \alpha_2 + \alpha_3 = 0$.

Problem 4. We have the following confidence ellipse for two regression parameters β_{15} and β_{75}. What can you comment, based on Figure 1, about the results of the following tests? Explain your reasons.

(1) Three null hypotheses denoted by H_0^1, H_0^2, and H_0^3, where
$H_0^1 : \beta_{15} = -0.4$,
$H_0^2 : \beta_{75} = -0.4$, and
$H_0^3 : \beta_{15} = \beta_{75} = -0.4$.
Answer: H_0^1: not reject; H_0^2: not reject; H_0^3: not reject.

(2) Another three null hypotheses denoted by H_0^4, H_0^5, and H_0^6, where
$H_0^4 : \beta_{15} = -0.2$,
$H_0^5 : \beta_{75} = 0.6$, and
$H_0^6 : \beta_{15} = -0.2, \text{ and } \beta_{75} = 0.6$.
Answer: H_0^1: not reject; H_0^2: reject; H_0^3: not reject.

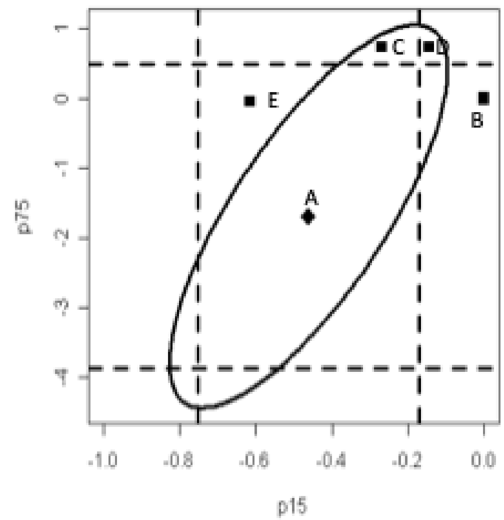

Figure 1. A confidence ellipse

(3) Now modify Figure 1 to Figure 2. Where should the point (-0.6,0) located if the data reject H_0^7 and H_0^8 but do not reject H_0^9, where
$H_0^7 : \beta_{15} = -0.6$,
$H_0^8 : \beta_{75} = 0$, and
$H_0^9 : \beta_{15} = -0.6, \text{ and } \beta_{75} = 0$.
Answer: D

Problem 5. In an Enchanted Kingdom, the Great Fox discovered a Six-Sided "Transitive" dice with numbers 1, 2, 3, 4, 5 and 6, each of which independently occurs with 1/6 probability whenever the dice is rolled.

- Each time she rolls the dice, she receives as many gold ingots as the number on the dice.
- She continues to roll the dice as long as a 1, 3 , 4 or 5 occur.
- But once a 2 or a 6 occurs, after she gets her gold ingots from that roll the dice vanishes and she never rolls the dice again.

She keeps rolling the dice until it vanishes.

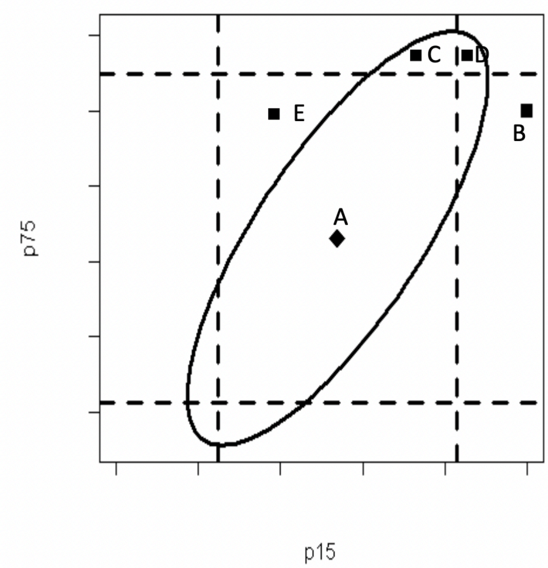

FIGURE 2. A modified confidence ellipse

The table below shows results of her up to first three rolls:

IN ALL ROLLS SHE GETS AND KEEPS THE NUMBER OF GOLD INGOTS SHOWN ON THE DICE			
First Roll Result	Second Roll Result	Third Roll Result	Forth, Fifth, Six, Seventh, and Succeding Results
2 or 6: Die vanishes	NO ROLL	NO ROLL	NO ROLL
1, 3, 4, 5: She rolls again	2 or 6: Die vanishes	NO ROLL	NO ROLL
	1, 3, 4, 5: She rolls again	2 or 6: Die vanishes	NO ROLL
		1, 3, 4, 5: She rolls again	And So On Until the Die Vanishes

(1) What is the expected number of times she rolls the dice before it vanishes?

Dice roll result	*Unconditional probability*
1st roll 6 - dice vanishes	$1/6$
1st roll 4, 2nd roll 2 - dice vanishes	$1/6 \times 1/6 = 1/36$
1st roll 3, 2nd roll 1, 3rd roll 2 - dice vanishes	$1/6 \times 1/6 \times 1/6 = 1/216$
1st roll 1, 2nd roll 3, 3rd roll 2 - dice vanishes	$1/6 \times 1/6 \times 1/6 = 1/216$
1st roll 1, 2nd roll 1, 3rd roll 1, 4th roll 1, 5th roll 2 - dice vanishes	$1/6 \times 1/6 \times 1/6 \times 1/6 \times 1/6 = 1/7776$

Answer: Number of rolls have Geometric distribution with 2/3 probability to continue rolling each time the expected number is $1/(1-2/3) = 3$.

(2) What is the expected number of gold Ingots that she acquires from all of her rolls?
Answer: ANSWER On her last roll where dice vanishes the expected number is (2 + 6) / 2 = 4
But she has an expectation of (3-1) = 2 other rolls. On each of those rolls her expected number of gold ingots is $(1+3+4+5)/4 = 3.25$.
So the Total Expected Number is $4 + 2 \times 3.25 = 10.5$.

(3) i) What possible combination of results on the 1st, 2nd, ... and so on dice rolls would give the Great Fox 6 gold ingots before the dice vanishes and ii) What is the Probability for each of these combination of Rolls? Remember the dice vanishes whenever a 2 or a 6 is rolled and the Fox receives her gold ingots from the roll before the dice vanishes.
Answer: These are the possible combinations that result in 6 gold ingots before the dice vanishes.

(4) Given you know that the Great Fox received 6 gold ingots before the dice vanished, from your answer to c what is the probability that the dice vanished on the second roll?
Answer: From the above table, the conditional probability is $(1/36)/(1/6 + 1/36 + 1/216 + 1/216 + 1/7776) = 216/63757 = 0.0034$.

Problem 6. Rutgers scientists fit a linear regression model to rates of crystallization of two substances they recently discovered in outer space; Noops and Poons based on Degrees Celsius. They had the following results on 10 independent observations (5 each for Poons and

X	Y_1	$\hat{y}_1$	$(Y_1 - \hat{y}_1)^2$
-2	*2*	*2.4*	*0.16*
-1	*4*	*3.2*	*0.64*
0	*4*	*4*	*0*
1	*4*	*4.8*	*0.64*
2	*6*	*5.6*	*0.16*

Noops).

$X =$ Degrees Celcius	Crystallization Rates	
	$Y_1 =$ for Noops	$Y = 2$ for Poons
-2	2	6
-1	4	4
0	4	4
1	4	4
2	6	2

Assume for each substance a linear model $Y = a + bX + e$ with e iid $N(0, \sigma^2)$.

(1) Calculate the least squares estimates for the slopes $\hat{b}_1$ for Noops and $\hat{b}_2$ for Poons.
Answer: The general formula is $\hat{b} = (\sum XY - (n\bar{x}\bar{y}))/(\sum X^2 - (n\bar{x}^2))$. *The same values of* X *were used for both Noops and Poons, and* $\bar{x} = (-2 + -1 + 0 + 1 + 2)/5 = 0$, $\sum X^2 = 4 + 1 + 0 + 1 + 4 = 10$. *For Noops,* $\bar{y}_1 = (2 + 4 + 4 + 4 + 6)/5 = 4$, $\sum XY_1 = -4 + -4 + 0 + 4 + 12 = 8$, *and so* $\hat{b}_1 = (\sum XY - (n\bar{x}\bar{y}))/(\sum X^2 - (n\bar{x}^2)) = (8 - 5 \times 0^2)/(10 - 5 \times 0 \times 10) = 0.8$. *For Poons,* $\bar{y}_2 = (6+4+4+4+2)/5 = 4$, $\sum XY_2 = -12 + -4 + 0 + 4 + 4 = -8$, *and so* $\hat{b}_2 = (\sum XY - (n\bar{x}\bar{y}))/(\sum X^2 - (n\bar{x}^2)) = (-8 - 5 \times 0^2)/(10 - 5 \times 0 \times 10) = -0.8$.

(2) What are $\hat{a}_1$ for Noops and $\hat{a}_2$ for Poons?
Answer: The general formula is $\hat{a} = \bar{y} + \hat{b}(0 - \bar{x})$. *For both Noops and Poons* $\bar{x} = 0$ *so* $\hat{b}(0 - \bar{x}) = 0$ *and* $\bar{y}_1 = \bar{y}_2 = 0$. *So for both Noops and Poons,* $\hat{a} = \bar{y} + \hat{b}(0 - \bar{x})4 + \hat{b}(0) = 4$.
NOTE ONLY ANSWER QUESTIONS 3 AND 4 FOR NOOPS.

(3) Estimate $\hat{\sigma}$ for Noops.
Answer: The general formula is $\hat{\sigma}^2 = \sum(Y - \hat{y})^2/(n-p)$ *where* n *is teh number of observations and* p *is teh number of parameters in the model. For both Noops and Poons,* $n = 5$ *and* $p = 2$.
From the table, $\sum(Y - \hat{y})^2 = 1.6$, *and* $\hat{\sigma}_1^2 = \sum(Y - \hat{y})^2/(n - p) = 1.6/3 = 0.533$.

(4) What is the estimated variance for $\hat{b}_1$ for Noops?

X	Y_1	$\hat{y}_1$	$(Y_1 - \hat{y}_1)^2$
-2	2	2.4	0.16
-1	4	3.2	0.64
0	4	4	0
1	4	4.8	0.64
2	6	5.6	0.16

Answer: The general formula is $\sum(Y - \hat{y})^2/(n - p)$ *where* n *is the number of observations and* p *is the number of parameters in the model. For both Noops and Poons,* $n = 5$ *and* $p = 2$.

From the table, $\sum(Y - \hat{y})^2 = 1.6$, *and* $\hat{\sigma}_1^2 = \sum(Y - \hat{y})^2/(n - p) = 1.6/3 = 0.533$.

(5) What is the estimated variance for $\hat{b}_1$ for Noops?
Answer: The general formula is $Var(\hat{b}) = \hat{\sigma}^2/\sum(X - \bar{x})^2$. *From* C *we have* $\hat{\sigma}^2 = 0.533$.

And as has already been calculated before in a different way up above $\sum(X - \bar{x})^2 = (-2 - 0)^2 + (-1 - 0)2 + (0 - 0)^2 + (1 - 0)^2 + (2 - 0)^2 = 10$. *So estimate* $Var(\hat{b}_1) = 0.0533/10 = 0.053$.